Ontario Peak Demand & Storage Adequacy Study

Analysis of IESO public data for 2025, with growth scenarios from the 2026 Annual Planning Outlook. Report generated automatically by run_study.py.

Key findings

- Ontario's 2025 peak was **24,862 MW** on June 24 in the hour ending 19:00 EST, matching the figure IESO reports for the year. All 100 highest-demand hours were in summer.
- **The peak has moved later in the day.** The median annual peak fell in the hour ending 16:00 EST over 2003-2012, versus 17:00 over 2020-2025, and 19:00 in 2025. Later peaks get less help from solar.
- **Storage hits diminishing returns early.** Up to **1.50 GW** of 4-hour storage lowers the 2025 net peak by its full rating. At 3 GW it lowers the peak by 1,968 MW (66% of rating), and at 5 GW by 2,617 MW (52%). 8-hour storage keeps full value up to 2.75 GW.
- **Solar and storage complement each other.** Adding 2.5 GW of solar narrows the evening peak, raising the value of 3 GW of 4-hour storage from 66% to 95% of its rating.
- **Storage relieves transmission stress but does not replace it.** On the IEEE 39-bus test system at 2035 load (+23%), a 500 MW battery at the best site (bus 3) cut N-1 contingencies with a violation from 38 to 21 and removed the 1 case with no load-flow solution, but the worst overload (line 21-22 outage, 160% loading) remains and needs a wires solution.

1. Data

All load and generation data come from IESO's public reports: the *Hourly Demand Report* (PUB_Demand_YYYY.csv, 2003-2025) and the *Generator Output and Capability Month Report* (PUB_GenOutputCapabilityMonth_YYYYMM.csv, all of 2025, 187 transmission-connected generators of 20 MW or more, with fuel type). The code downloads these from reports-public.ieso.ca and falls back to the bundled copies when offline.

Check against IESO's published figures: the data give a 2025 peak of 24,862 MW on June 24 and annual Ontario Demand of 145.6 TWh, the same as IESO's 2025 Year in Review (24,862 MW on June 24; 145.6 TWh).

Gaps: the bundled monthly generator files are missing 6 days (2025-03-27, 2025-03-28, 2025-03-29, 2025-03-30, 2025-03-31, 2025-06-30). None is near the peak (the highest demand on any of them is far below the annual peak), so the peak and storage results are unaffected; downloading from IESO fills them in. IESO's own 2025 demand file has no row for hour 1 of May 1, so that hour is skipped. Net load is Ontario Demand minus transmission-connected wind and solar output.

2025 transmission-connected output by fuel (TWh, over the hours with data): Nuclear 76.5, Hydro 36.3, Gas 31.9, Wind 13.9, Solar 0.7, Biofuel 0.3.

2. Peak demand and net load

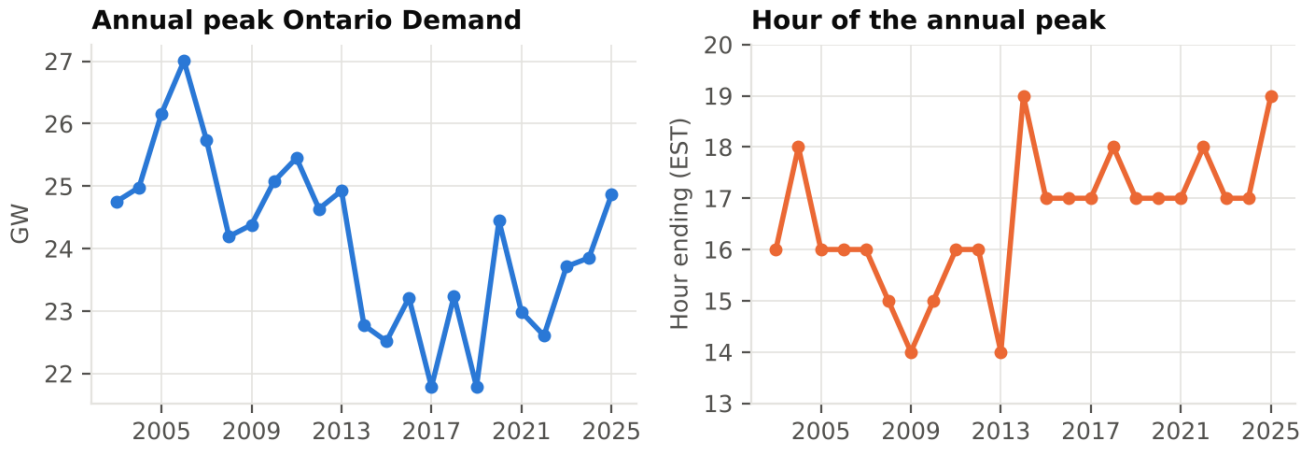


Figure 1. Annual peak Ontario Demand and the hour it occurred, 2003-2025, from IESO Hourly Demand Reports. Winter peaks: 2004, 2014; every other year peaked in summer.

At the 2025 peak hour, wind supplied 607 MW and transmission-connected solar 39 MW, so the net load peak was 24,216 MW. Only 19 hours were within 5% of the net peak and 145 within 10%. A short, sharp peak is what makes energy-limited storage effective, up to a point.

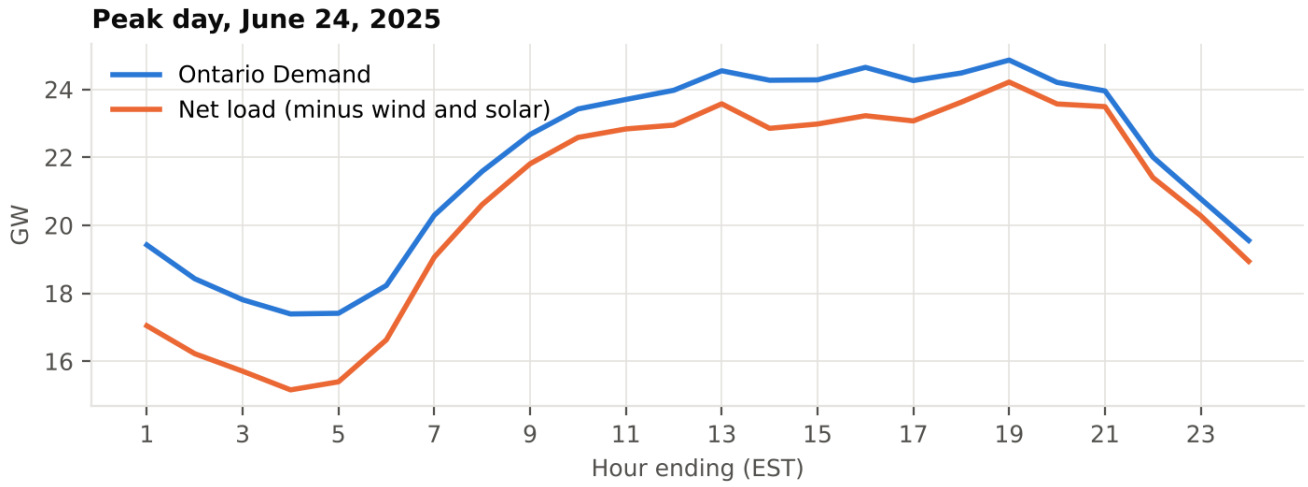


Figure 2. The peak day: demand and net load by hour.

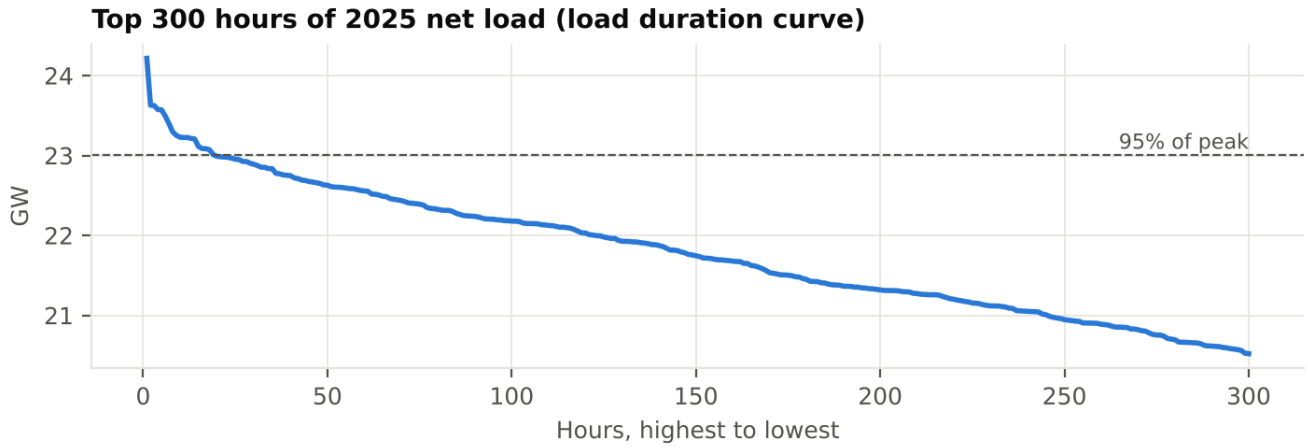


Figure 3. The 300 highest net-load hours of the year. The steep left end is the part storage can shave.

3. How much does storage lower the peak?

For every day, a storage fleet of power P and duration H discharges in the highest hours to bring that day's net load down as far as its power and energy allow, then recharges the same day in the lowest hours (round-trip efficiency 85%) without creating a new peak. The fleet's value is how far the annual peak falls. This assumes perfect foresight and a full battery every morning, so it is an upper bound.

Storage on 2025 net load

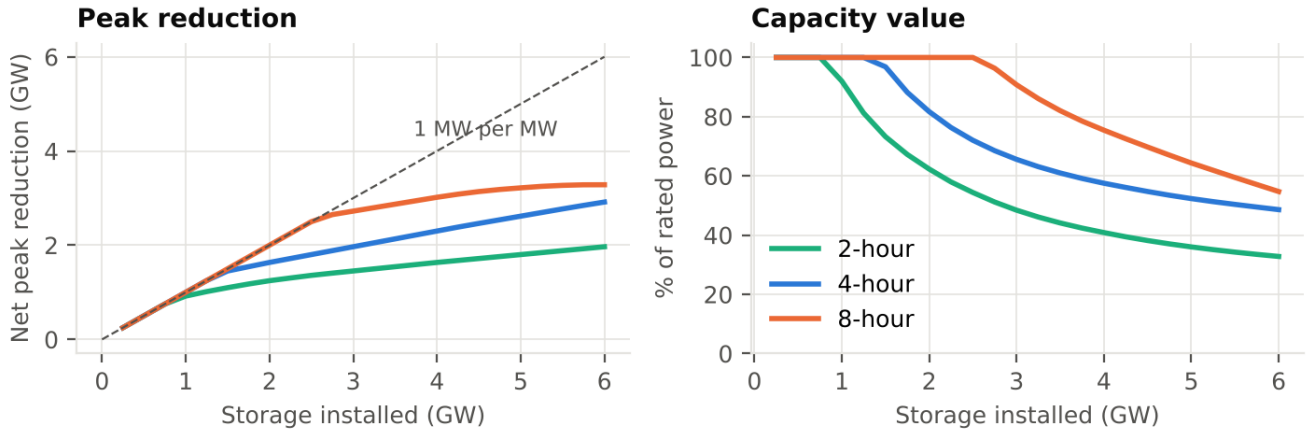


Figure 4. Peak reduction and capacity value for 2-, 4- and 8-hour storage on 2025 net load. The dashed line is full value (1 MW of peak reduction per MW installed).

Storage	1 GW	2 GW	3 GW	5 GW	Full value up to
2-hour	920 MW (92%)	1,246 MW (62%)	1,453 MW (48%)	1,801 MW (36%)	0.75 GW
4-hour	1,000 MW (100%)	1,634 MW (82%)	1,968 MW (66%)	2,617 MW (52%)	1.50 GW
8-hour	1,000 MW (100%)	2,000 MW (100%)	2,724 MW (91%)	3,218 MW (64%)	2.75 GW

Table 1. Net peak reduction (2025), with capacity value as a share of rated power.

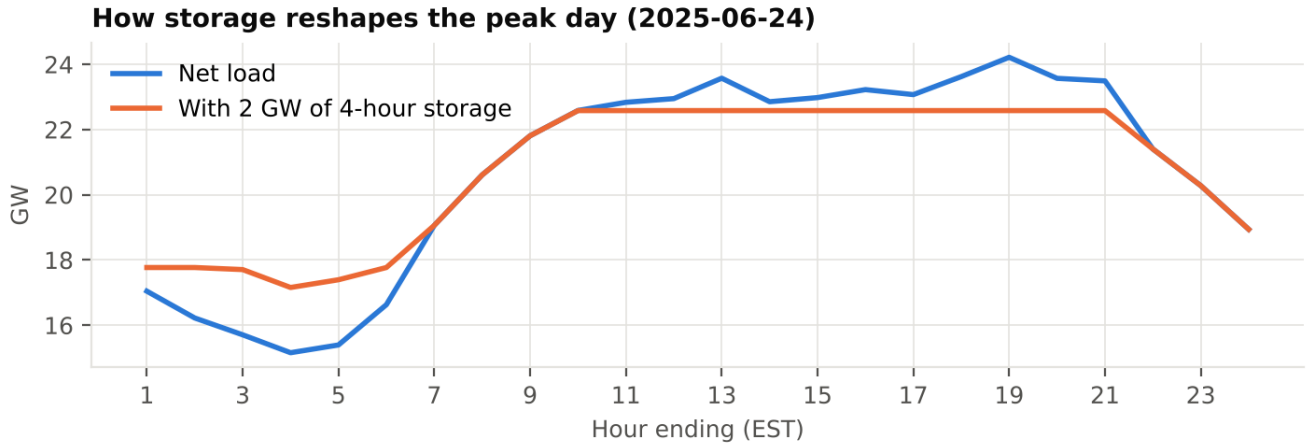


Figure 5. 2 GW of 4-hour storage on the peak day. The new annual peak is 22,582 MW: high load lasts from late morning to 9 pm, so the 8 GWh of energy only stretches far enough to trim the top of a long plateau.

Growth to 2035

Scaling 2025 demand by the APO reference growth rate (about 2.1% a year, a factor of 1.231 by 2035) raises the net peak to 29,959 MW. Because the shape is scaled uniformly, the pattern is the same but slightly larger: 4-hour storage keeps full value up to 1.75 GW and 8-hour storage up to 3.25 GW.

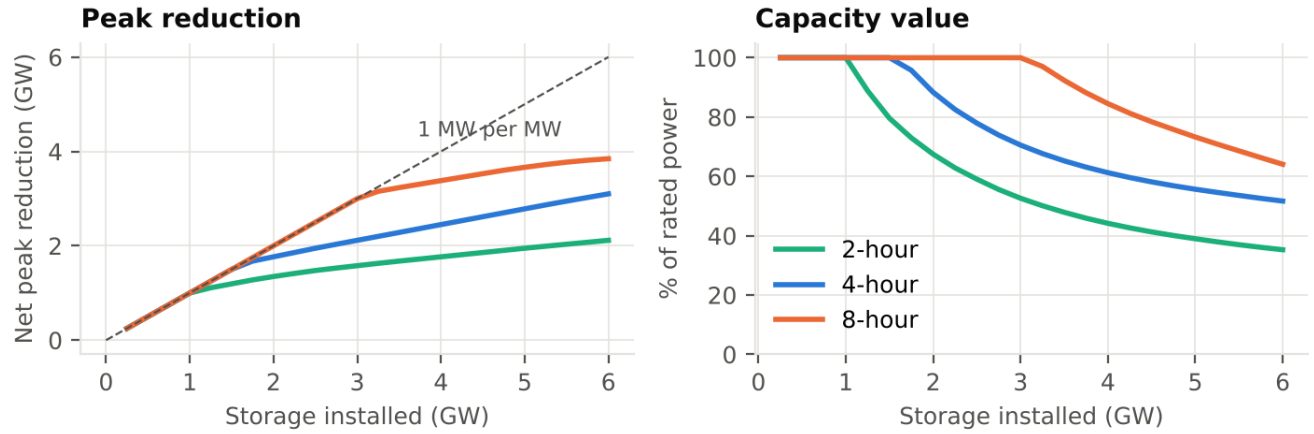


Figure 6. The same sweep on 2035 net load.

Solar sensitivity

Ontario's transmission-connected solar fleet is small (about 438 MW available at most in the IESO data). Adding solar with the same hourly profile pushes the highest net-load hours later and makes them narrower, so 4-hour storage covers more of them:

Extra solar	Net peak	Median hour of top 100 (HE, EST)	4h at 2 GW	4h at 3 GW	4h at 4 GW
+0 GW	24,216 MW	17.0	82%	66%	58%
+2.5 GW	23,993 MW	19.0	100%	95%	82%
+5 GW	23,771 MW	19.5	100%	100%	91%

Table 2. Effect of extra solar on the net peak and on 4-hour storage capacity value.

4. Transmission: N-1 contingency analysis

Ontario's detailed transmission model is not public, so this section uses the IEEE 39-bus test system in pandapower as a stand-in. Loads and generator dispatch are scaled by the APO growth factors for 2030 and 2035. Every line and network transformer is taken out one at a time (38 contingencies; generator step-up transformers are excluded because losing one is a generation contingency). A case is a violation if any branch exceeds 100% of its rating, any load bus falls below 0.95 p.u., or the load flow has no solution.

The battery (500 MW, about 6.5% of 2035 test-system load, similar to a few GW in Ontario) was tested at every load bus at 2035 load, and the site with the lowest severity score was chosen. In each scenario the battery is discharged at the level (0-100%) that gives the fewest problems, as an operator would.

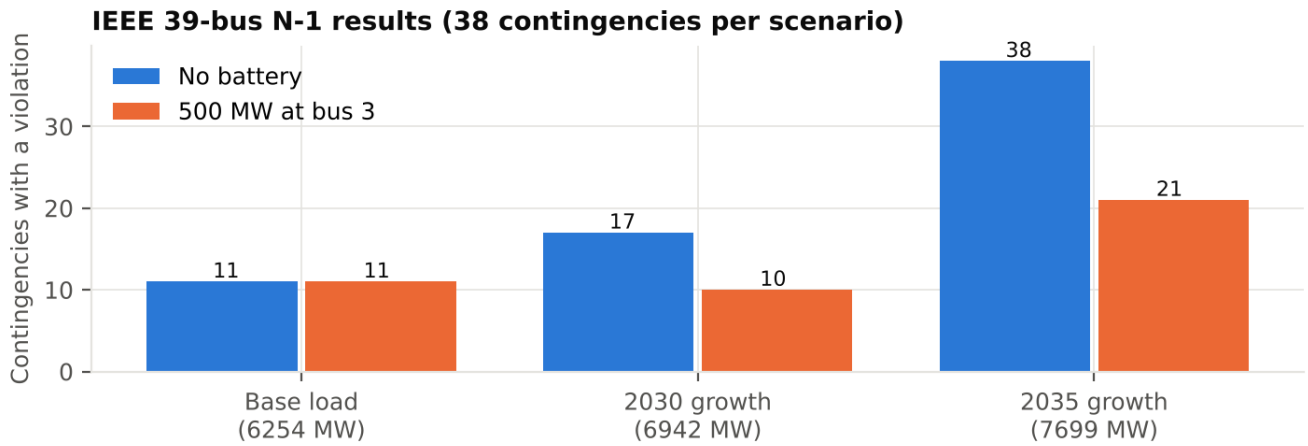


Figure 7. Contingencies with a violation, with and without the battery.

Load level	Battery	Output	Failing (of 38)	Over-load	Low V	No solu-tion	Worst load	Lowest V (p.u.)
Base load (6,254 MW)	none	0 MW	11	10	1	0	158%	0.937
Base load (6,254 MW)	500 MW at bus 3	0 MW	11	10	1	0	158%	0.937
2030 growth (6,942 MW)	none	0 MW	17	15	4	0	164%	0.907
2030 growth (6,942 MW)	500 MW at bus 3	375 MW	10	9	1	0	164%	0.917
2035 growth (7,699 MW)	none	0 MW	38	37	37	1	234%	0.825
2035 growth (7,699 MW)	500 MW at bus 3	500 MW	21	14	16	0	160%	0.876

Table 3. N-1 results by scenario.

Reading the results. The test system as published already has violations at base load, mostly the outage of line 21-22, which overloads a parallel path. Growth makes things much worse: at 2035 load, 38 of 38 contingencies fail, 37 with low voltage and 1 with no solution at all, a sign of voltage collapse. The battery at bus 3 brings 2030 back to 10 violations (from 17) using 375 MW, and 2035 to 21. It cannot fix the line 21-22 overload, which comes from the network's structure rather than total load, so that one needs a transmission upgrade. In base load the best choice is not to discharge at all.

Rank	Bus	Violating contingencies	Severity score
1	3	21	613
2	18	23	615
3	27	24	660
4	15	25	669
5	16	25	683

Table 4. Best five battery sites at 2035 load (IEEE bus numbers). Worst site: bus 31.

5. Assumptions and limitations

- Peak reduction is a simple stand-in for capacity value. IESO's resource adequacy work is probabilistic (how often supply falls short across many weather and outage conditions); a natural next step is an effective load-carrying capability (ELCC) calculation.
- Storage dispatch assumes perfect daily foresight and a full battery each morning, so results are an upper bound.
- Growth is a uniform scaling of 2025's hourly shape. The APO expects winter demand to grow faster with electrification, which would change the shape of the peak; IESO's zonal hourly forecast would capture that.
- Net load uses transmission-connected wind and solar only. Embedded (distribution-connected) solar already lowers the Ontario Demand that IESO measures.
- The network study uses a test system, not Ontario's grid, and the slack bus covers any generation shortfall.

Sources

- IESO Hourly Demand Report and Generator Output and Capability Month Report, reports-public.ieso.ca
- IESO, 2025 Year in Review (peak demand and annual energy), ieso.ca
- IESO, 2026 Annual Planning Outlook (reference scenario growth), ieso.ca
- IEEE 39-bus test system, via pandapower.networks.case39